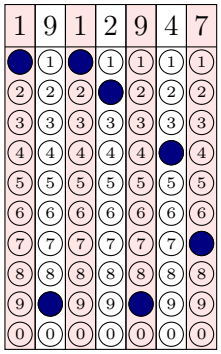
TEST VARIANTI RAQAMI: **1912947**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



1. $\dfrac{(x^2-9):(3+x)-2}{(x^2-16):(x-4)+2}=(x-1):(x+1)$ tenglamaning ildizini toping.

A) $x\in\varnothing$ B) $\frac{4}{5}$ C) $\frac{1}{9}$ D) $3,5$

2. $f(x)=3^{|x|}-2$ funksiyaning qiymatlar sohasini toping.

A) $(0;+\infty)$ B) $(-2;+\infty)$ C) $[-1;+\infty)$ D) $(-1;+\infty)$

3. $f(x)=kx+2$ funksiya k ning qanday qiymat(lar)ida juft funksiya bo‘ladi?

A) $k<0$ B) $k>0$ C) $k=2n,\ n\in N$ D) $k=0$

4. Hisoblang: $\sqrt{3-2\sqrt{2}}+\sqrt{3+2\sqrt{2}}$

A) $\sqrt{2}-1$ B) $\sqrt{2}+1$ C) $2\sqrt{2}$ D) $\sqrt{2}$

5. Sayyoh belgilangan yo‘lning $\frac{1}{6}$ qismini bosib o‘tgach, yo‘l yarmigacha yana 16 km qolgan bo‘lsa, belgilangan yo‘l uzunligini (km) toping.

A) *36* B) *42* C) *48* D) *54*

6. $ABCD$ trapetsiyaning AB katta asosida E nuqta olingan. CE kesma AD yon tomoniga parallel. Agar $AE{:}EB{=}3{:}5$ bo‘lsa, $AECD$ to‘rtburchak yuzini BCE uchburchak yuziga nisbatini toping.

A) $\frac{6}{5}$ B) $\frac{3}{2}$ C) $\frac{3}{5}$ D) $\frac{5}{6}$

7. $\dfrac{3}{\sqrt[3]{9x^2-1}+3}+\dfrac{2}{\sqrt[3]{9x^2-1}+2}=2$ tenglama nechta haqiqiy ildizga ega?

A) *0* B) *1* C) *2* D) *4*

8. Integralni hisoblang: $\int\limits_{-3}^{3}|x-2|dx$

A) *16,5* B) *9,5* C) *13* D) *6*

9. 0,6(8) son 0,(31) sondan necha marta katta?

A) *3,(3)* B) *2,2* C) *2* D) *2,(2)*

10. Ushbu $\left(\dfrac{a^{\frac{3}{2}}+b^{\frac{3}{2}}}{a-b}-\dfrac{a-b}{a^{\frac{1}{2}}+b^{\frac{1}{2}}}\right)\cdot\dfrac{a-b}{\sqrt{ab}}-\sqrt{a}$ ifodaning $a=\sqrt{2}$, $b=8$ dagi qiymatini toping.

A) *0* B) *2* C) $-\sqrt{2}$ D) $2\sqrt{2}$

11. Hisoblang: $10\cdot\left(3^{\log_{\sqrt[3]{4}}4}-7^{2\log_{343}8}+12\right):7^{\log_{49}196}$

A) 20 B) 42 C) 35 D) 25

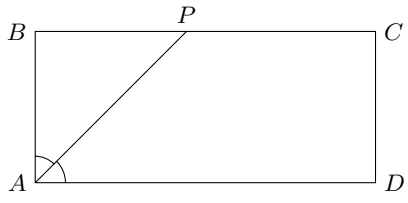
12. Markazlari har xil nuqtalarda bo‘lgan 3 ta aylana ko‘pi bilan nechta nuqtada kesishadi?

A) *4* B) *6* C) *7* D) *3*

13. Umumiy hadi $x_n=\dfrac{3n}{2}+1$ formula bilan berilgan ketma-ketlikning dastlabki yigirmata hadining o‘rta arifmetigini toping

A) $16\frac{1}{4}$ B) $14\frac{3}{4}$ C) $16\frac{3}{4}$ D) $14\frac{1}{4}$

14. Rasmda $ABCD$ to‘g‘ri to‘rtburchak, BAD burchakning AP bissektrisasi tasvirlangan. Agar $BP{=}4$ va $PC{=}5$ bo‘lsa, $APCD$ trapetsiyaning yuzini toping.



A) *24* B) *28* C) *27* D) *25*

15. Agar prizmaning qirralar soni yoqlari sonidan 24 taga ko‘p bo‘lsa, prizmaning diagonallari soni uchlari sonidan nechtaga ko‘p?

A) *117* B) *91* C) *130* D) *104*

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16. $\sin x - \sin\dfrac{3x}{2}\cos\dfrac{x}{2} = 0$ tenglamaning $[0;\pi]$ kesmadagi turli yechimlari yig‘indisini toping.

A) $\dfrac{\pi}{3}$ B) π C) $\dfrac{4\pi}{3}$ D) $\dfrac{5\pi}{6}$

17. Uchburchakli prizmaning yon qirralari orasidagi masofalar 37 cm, 15 cm va 26 cm ga teng. Agar prizmaning yon qirrasi uzunligi 5 cm bo‘lsa, uning hajmini (cm^3) toping.

A) *770* B) *760* C) *790* D) *780*

18. $\dfrac{1}{3}$ va $\dfrac{5}{6}$ sonlar orasida maxraji 24 bo‘lgan, qisqarmaydigan barcha kasrlar yig‘indisini toping.

A) *2,5* B) *1,5* C) $4\dfrac{1}{6}$ D) *5,5*

19. $0,009 \cdot 0,02 \cdot 10^6$ ko‘paytmani standart shaklga keltiring.

A) $1,8 \cdot 10$ B) $1,8 \cdot 10^{-1}$ C) $1,8 \cdot 10^{-2}$ D) $1,8 \cdot 10^{2}$

20. $\left(x^2 - 2x - 24\right)\sqrt{10x - x^2} < 0$ tengsizlikning eng katta butun yechimini toping.

A) *6* B) *7* C) *5* D) *9*

21. a ning qanday qiymatlarida uzunliklari mos ravishda $a+2$; 4 va $2a-1$ bo‘lgan kesmalardan uchburchak yasash mumkin?

A) $(0;7)$ B) $(1;7)$ C) $(0;5)$ D) $(1;8)$

22. Ko‘paytuvchilarga ajrating:
$(3a+2b)^2 - (2a+3b)^2$

A) $(a+5b)\cdot(a-b)$ B) $5(a+b)\cdot(a-b)$
C) $(5a+b)\cdot(a-b)$ D) $-5(a+b)\cdot(a-b)$

23. $y = 3x^2 - 6x + 7$ kvadrat funksiyaning $(0;\,0)$ nuqtaga nisbatan simmetrik funksiyasini aniqlang.

A) $y = 3x^2 + 6x + 7$ B) $y = -3x^2 - 6x - 7$
C) $y = 3x^2 - 6x + 7$ D) $y = -3x^2 + 6x - 7$

24. Hisoblang:
$139 \cdot 163 - 160 \cdot 139 + 141 \cdot 175 - 172 \cdot 141$

A) *852* B) *840* C) *870* D) *864*

25. Uchlari $A(3;6)$, $B(5;12)$ va $C(9;4)$ nuqtalarda bo‘lgan uchburchakning BD medianasi bo‘yicha yo‘nalgan birlik vektorni toping.

A) $\left(\dfrac{3}{5};-\dfrac{4}{5}\right)$ B) $\left(\dfrac{1}{5\sqrt{2}};\dfrac{7}{5\sqrt{2}}\right)$
C) $\left(\dfrac{1}{5\sqrt{2}};-\dfrac{7}{5\sqrt{2}}\right)$ D) $\left(-\dfrac{3}{5};\dfrac{4}{5}\right)$

26. Agar qisqarmaydigan kasrning surati 3 ga orttirilsa, kasrning qiymati $\dfrac{6}{7}$ ga, maxraji 2 ga kamaytirilsa, kasrning qiymati $\dfrac{3}{4}$ ga teng bo‘ladi. Berilgan kasrning $\dfrac{7}{27}$ qismini toping.

A) $\dfrac{3}{4}$ B) $\dfrac{2}{3}$ C) $\dfrac{1}{12}$ D) $\dfrac{1}{6}$

27. $\left(\dfrac{1}{\sin^2\alpha} - \dfrac{1}{\cos^2\alpha}\right)\cdot\dfrac{\sin 2\alpha\,\mathrm{tg}\,\alpha}{1 - tg^2\alpha}$ ifodani soddalashtiring.

A) 2 B) $\sin\alpha$ C) $\cos\alpha$ D) 1

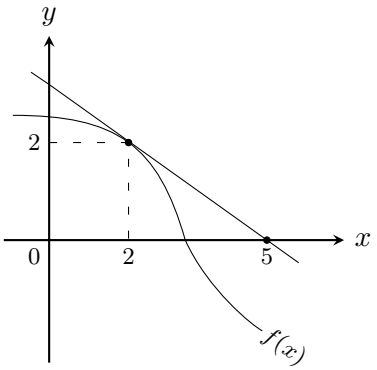
28. $A = \{a;\ b;\ c;\ d;\ e;\ f\}$ to‘plamning nechta qism to‘plamlarida, b elementi bo‘lib, c elementi qatnashmaydi?

A) *28* B) *32* C) *16* D) *8*

29. $f(x) = \ln(2x+5)$ funksiyaning x_0=2 nuqtadagi hosilasini toping.

A) $\dfrac{2}{9}$ B) $\dfrac{4}{9}$ C) 0 D) $-\dfrac{2}{9}$

30. Rasmda $y = f(x)$ funksiya grafigi va unga $(2;2)$ nuqtadan o‘tkazilgan urinmasi tasvirlangan. Agar $g(x) = \left(x^2-2\right)\cdot f(x)$ bo‘lsa, $g'(2)$ ni toping.



A) $9\dfrac{2}{3}$ B) $9\dfrac{1}{3}$ C) $6\dfrac{1}{3}$ D) $6\dfrac{2}{3}$

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Test varianti “Test topshiriqlari to‘plami 2019” asosida shakllantirilgan:

1 – 59-bet 8	7 – 66-bet 73	13 – 31-bet 12	19 – 24-bet 21	25 – 118-bet 40
2 – 54-bet 25	8 – 89-bet 33	14 – 91-bet 15	20 – 69-bet 32	26 – 22-bet 51
3 – 71-bet 11	9 – 11-bet 9	15 – 107-bet 7	21 – 98-bet 60	27 – 36-bet 18
4 – 28-bet 58	10 – 51-bet 62	16 – 44-bet 89	22 – 47-bet 22	28 – 119-bet 2
5 – 19-bet 32	11 – 55-bet 4	17 – 112-bet 48	23 – 75-bet 59	29 – 78-bet 16
6 – 102-bet 91	12 – 123-bet 5	18 – 15-bet 55	24 – 9-bet 29	30 – 81-bet 41

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